Microbiology

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MICROBIOLOGY

1. BACTERIA
Prokaryotic
Shapes: cocci, bacilli, spirilla
Gram +ve: thick peptidoglycan (purple)
Gram -ve: thin + outer membrane (pink)
Reproduction: binary fission
Pathogenic: TB, cholera, tetanus
Beneficial: gut flora, nitrogen fixation

2. VIRUSES
Acellular
DNA or RNA + protein coat
Obligate intracellular parasites
Lytic vs Lysogenic
Examples: HIV, Influenza, COVID-19, Herpes

3. FUNGI
Eukaryotic
Yeasts (unicellular), Molds (multicellular)
Cell wall: chitin
Reproduction: spores
Examples: Candida, Aspergillus
Beneficial: Penicillin, yeast in bread

4. PROTOZOA
Unicellular eukaryotes
Malaria (Plasmodium)
Amoeba, Paramecium

5. IMMUNITY
Innate: skin, mucus, phagocytes
Adaptive: B cells, T cells
Vaccines: attenuated, inactivated, subunit
Antibiotics: cell wall, protein synthesis

6. CULTURE
Nutrient agar
Streak plate
Incubation 37 C
Sterilization: autoclave